Retrieval-Augmented Intelligent Legal Assistant for Court Judgments: A Hybrid Retrieval and Evidence-Grounded Generation Approach

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ABSTRACT

Court judgments are lengthy, citation-dense, and written in domain-specific legal language, which makes keyword-based search prone to missing semantically related precedents and makes unconstrained large language models (LLMs) prone to producing unsupported or hallucinated legal claims. This paper proposes a Retrieval-Augmented Generation (RAG) system for court-judgment question answering that combines lexical retrieval (BM25), dense semantic embeddings, metadata filtering, and cross-encoder reranking to build a hybrid retrieval pipeline, followed by legal-structure-aware chunking and an evidence-first generation stage in which every material claim in the generated answer is required to be traceable to a retrieved passage. A verification module performs a faithfulness check and triggers a low-confidence "insufficient evidence" response when supporting evidence is weak, rather than allowing the system to answer regardless of relevance. We describe the proposed architecture, the dataset and preprocessing strategy, the mathematical formulation of the hybrid scoring and evaluation metrics, and a five-dimension evaluation protocol spanning retrieval quality (Recall@k, Precision@k, MRR/nDCG), answer quality, faithfulness, usability (latency, citation coverage), and component-level ablation. Because this work is reported at the design and prototype-development stage, Section 5 presents an illustrative walkthrough of the evaluation framework using representative example outputs rather than final experimental results; the complete quantitative evaluation is scheduled for completion at the end of the ten-week project timeline described in Section 4. The proposed system is intended as a research-support tool that augments, rather than replaces, human legal judgment, and is positioned to reduce unsupported claims and improve citation traceability relative to keyword-only legal search engines and standalone LLMs.

Keywords

Legal Natural Language Processing; Retrieval-Augmented Generation; Hybrid Retrieval; Court Judgment Question Answering; Explainable AI; Faithfulness Evaluation; Cross-Encoder Reranking; Legal Information Retrieval

1. INTRODUCTION

1.1 Background

Legal research depends on the ability to locate, interpret, and cite relevant precedent quickly and accurately. Court judgments, however, are long, procedurally structured, and rich in cross-references to statutes and prior cases, which makes them difficult to search using conventional keyword or Boolean methods. At the same time, general-purpose large language models have shown strong fluency in answering natural-language questions but are known to generate confident-sounding statements that are not actually grounded in any source document, a failure mode that is unacceptable in a legal context where every claim must be traceable to an authoritative text. Text analytics techniques — including information retrieval, sentence embeddings, semantic similarity, reranking, and retrieval-augmented generation — offer a way to combine the recall benefits of statistical search with the language-understanding benefits of neural models, while constraining generation to retrieved evidence.

1.2 Problem Statement

Court judgments are long, citation-heavy, and domain-specific documents. Keyword search can miss semantically related precedents because it depends on exact term overlap rather than legal meaning, while standalone LLMs used for legal question answering may produce unsupported or hallucinated legal claims because they generate text from parametric knowledge rather than from verifiable source passages. There is therefore a need for a system that retrieves legally relevant passages with high recall and precision, and that generates answers whose claims are explicitly and verifiably linked to retrieved evidence, with the ability to abstain when the evidence is insufficient.

1.3 Research Gap

Existing legal search tools and prior academic work on legal judgment prediction and legal retrieval-augmented generation tend to address retrieval, prediction, or generation in isolation, and do not consistently evaluate retrieval quality, answer correctness, and faithfulness together within a single end-to-end pipeline for court-judgment question answering (elaborated in Section 2.6).

1.4 Research Objectives

1. To build a searchable corpus of court judgments with hybrid retrieval — BM25, dense embeddings, and metadata filters — refined by cross-encoder reranking.

2. To develop legal-aware retrieval that combines dense and lexical signals and attaches case, paragraph/page, and citation-level evidence to every retrieved item.

3. To generate concise answers grounded in retrieved passages, and to evaluate retrieval quality, answer quality, and faithfulness of the generated answers.

4. To demonstrate a working prototype for legal research assistance — explicitly positioned as a decision-support aid rather than an autonomous legal decision-making system.

1.5 Research Questions

- RQ1: Does hybrid retrieval (lexical + dense + metadata filtering, refined by reranking) retrieve relevant precedents and passages more effectively than keyword-only or dense-only retrieval?
- RQ2: Does legal-structure-aware chunking (around issues, holdings, statutes, and cited precedents) improve retrieval precision compared to fixed-size chunking?

- RQ3: Does evidence-first, citation-linked generation reduce unsupported or hallucinated claims relative to unconstrained LLM generation?
- RQ4: Which pipeline component (lexical retrieval, dense retrieval, or reranking) contributes most to overall answer quality and faithfulness?
- RQ5: Is the resulting system practical for research use in terms of latency, citation coverage, and answer length?

1.6 Contributions

- Innovation: A hybrid retrieval design combining BM25/keyword search, dense embeddings, metadata filters, and cross-encoder reranking for the court-judgment domain.
- Creativity: Legal-structure-aware chunking that segments judgments around case sections, issues, holdings, statutes, and cited precedents rather than using fixed-size windows.
- Novelty: An evidence-first answer-generation strategy in which each claim is linked to retrieved passages, and unsupported claims trigger a low-confidence or abstain response instead of a fabricated answer.
- Research contribution: A multi-dimensional evaluation matrix that separately measures Recall@k / MRR, citation correctness, groundedness (faithfulness), answer relevance, and latency, rather than relying on a single aggregate score.

1.7 Organization of the Paper

The remainder of this paper is organized as follows. Section 2 reviews existing legal search approaches, machine learning and deep learning approaches to legal text analytics, and explainability techniques relevant to retrieval-augmented generation, and positions the identified research gap. Section 3 presents the proposed methodology, including the overall framework, dataset, preprocessing, feature engineering, the proposed hybrid retrieval-and-generation model, baseline models, the explainability method, and the mathematical formulation of scoring and evaluation. Section 4 describes the planned experimental setup. Section 5 presents an illustrative evaluation walkthrough. Section 6 discusses the interpretation and implications of the findings. Section 7 states the limitations of the current stage of work, and Section 8 concludes with directions for future work.

2. RELATED WORK

2.1 Existing Approaches

Most operational legal research platforms rely on keyword or Boolean search over full judgment texts. These systems return a ranked list of whole documents or fixed pages that the user must read individually to locate the relevant holding, and they provide no confidence signal about whether a returned document is actually relevant to the query. Because they match on surface-level terms rather than legal meaning, they can miss precedents that are conceptually related but lexically dissimilar, and they can split holdings and exceptions across page boundaries.

2.2 Machine Learning Approaches

Zhang and Dou (2023) show that incorporating historical case retrieval improves the realism of legal judgment prediction, supporting precedent retrieval as a first-class component of a legal reasoning

pipeline rather than an auxiliary step. Traditional statistical retrieval methods such as BM25 (Robertson & Zaragoza, 2009) remain a strong lexical baseline for legal text, particularly for exact statutory or citation matches, and are commonly combined with machine-learned ranking methods to improve precision.

2.3 Deep Learning Approaches

Neural approaches to retrieval and generation have substantially improved semantic matching over purely lexical methods. Dense Passage Retrieval (Karpukhin et al., 2020) and Sentence-BERT (Reimers & Gurevych, 2019) enable semantic similarity search using learned embeddings rather than exact term overlap, and cross-encoder re-ranking with transformer models such as BERT (Devlin et al., 2019; Nogueira & Cho, 2019) has been shown to substantially improve the precision of an initial retrieval candidate set. Retrieval-Augmented Generation (Lewis et al., 2020) formalizes the combination of a retriever with a generator so that language-model outputs are conditioned on retrieved evidence rather than generated purely from parametric knowledge, and domain-adapted legal language models such as LEGAL-BERT (Chalkidis et al., 2020) and Lawformer (Xiao et al., 2021) demonstrate that legal-domain pretraining improves downstream legal NLP tasks. Nigam et al. (2024) report that adding legal information improves legal judgment prediction, while noting that reliability at an expert-usable level remains an open problem — directly motivating the evidence-grounding approach adopted in this paper.

2.4 Explainability / Existing Techniques

Explainability in this domain concerns whether a generated legal answer can be traced back to a specific, verifiable source. Pipitone and Houir Alami (2024), through the LegalBench-RAG benchmark, find that precise, narrowly scoped passage snippets are preferable to large, imprecise chunks for legal RAG, which motivates fine-grained, legally structured chunking rather than fixed-size windows. Model-agnostic explanation techniques such as LIME (Ribeiro, Singh, & Guestrin, 2016) and SHAP (Lundberg & Lee, 2017) provide general-purpose feature-attribution explanations, but for generative legal question answering, claim-level citation attachment and entailment-style faithfulness scoring (Shuster et al., 2021; Ji et al., 2023) are more directly applicable because they explain individual generated claims rather than a single model prediction. Nigam et al. (2025), in NyayaRAG, combine realistic case context with precedent retrieval for legal judgment prediction under the Indian common-law system, motivating an India-relevant retrieval-and-explanation pipeline design.

2.5 Critical Comparison

Table 1 summarizes how the proposed system differs from existing keyword-based legal search platforms across the dimensions most relevant to evidence-grounded legal question answering.

Dimension	Existing Legal Search Websites	Proposed RAG Assistant
Search method	Keyword / Boolean search over full judgments; can miss semantically related precedents.	Hybrid retrieval — BM25 + dense embeddings + metadata filters, refined by cross-encoder reranking.
Document handling	Returns whole judgments or fixed pages; holdings and exceptions can be split apart.	Legal-structure-aware chunking around issues, holdings, statutes, and precedents.
Answer format	A ranked list of documents; the user must read each judgment to find the answer.	A concise, generated answer grounded directly in the retrieved passages.

Dimension	Existing Legal Search Websites	Proposed RAG Assistant
Evidence & citations	No claim-level links; citations must be located manually inside the judgment.	Every claim is linked to its case, paragraph/page, and citation.
Reliability check	No confidence signal; results are shown regardless of relevance.	Faithfulness check with a low-confidence / abstain response when evidence is weak.

Table 1. Existing keyword-based legal search versus the proposed hybrid RAG assistant.

Table 2 summarizes the literature reviewed and its implication for the design choices made in Section 3.

Work	Key Finding	Implication for This Project
Zhang & Dou (2023)	Historical case retrieval improves realism of legal judgment prediction.	Treat precedent retrieval as a first-class pipeline component.
Nigam et al. (2024)	Additional legal information improves prediction; expert-level reliability remains open.	Ground responses in statutes/precedents; avoid overclaiming.
Pipitone & Houir Alami (2024)	Precise snippets outperform large, imprecise chunks in legal RAG.	Optimize passage-level retrieval and legal-structure-aware chunking.
Nigam et al. (2025)	Combines realistic case context with precedent retrieval.	Motivates an India-focused retrieval-plus-explanation pipeline.
Wiggers & Hartz (2024)	Proposes measures of legal relevance specific to RAG applications.	Adopt legal-relevance-aware evaluation, not only generic IR metrics.
Magesh et al. (2024)	Leading AI legal research tools still produce unsupported or incorrect claims.	Motivates an explicit faithfulness check and abstain mechanism.

Table 2. Literature survey and design implications.

2.6 Research Gap

- Retrieval gap: Generic semantic search may retrieve topically similar but legally irrelevant text, without regard to legal hierarchy or temporal validity.
- Chunking gap: Fixed-size chunks can split holdings, exceptions, citations, and procedural context across boundaries.
- Evidence gap: A fluent answer is not sufficient; users need claim-level provenance and direct judgment evidence.
- Evaluation gap: Legal RAG requires separate measurement of retrieval relevance, answer correctness, and faithfulness, rather than a single blended score.
- Scope gap: Many prior studies address prediction or retrieval in isolation; an end-to-end court-judgment assistant must connect ingestion, retrieval, and grounded response generation.

3. PROPOSED METHODOLOGY

3.1 Overall Framework

The proposed system follows a six-stage end-to-end pipeline for court-judgment question answering: (1) Corpus — ingest judgments and associated metadata; (2) NLP Preparation — OCR and text cleaning; (3) Indexing — build a BM25 lexical index and a dense vector index; (4) Retrieval — combine hybrid top-k retrieval with cross-encoder reranking; (5) Generation — produce an answer with an LLM conditioned on the retrieved evidence; and (6) Verification — attach citations and run a faithfulness check before the answer is returned to the user. Given a user query such as "Which precedents support X?", the system returns an answer together with citations and a confidence indicator, rather than a bare document list.

3.2 Dataset

The corpus consists of publicly accessible court-judgment repositories or institution-provided judgment datasets, together with associated case metadata (court, date, jurisdiction, case type, and citation identifiers). To control scope during initial development, the corpus is deliberately limited to a single jurisdiction or court before being extended, which keeps legal terminology and citation conventions consistent within the initial corpus.

3.3 Data Preprocessing

Raw judgments are converted to clean text through OCR (for scanned or image-based sources) and text-cleaning steps that normalize whitespace, remove artefacts such as page headers/footers and stamps, and standardize citation formatting so that case and statute references can be reliably parsed downstream.

3.4 Feature Engineering

Rather than splitting judgments into fixed-size, arbitrary windows, the pipeline performs legal-structure-aware chunking that segments each judgment around its constituent sections — facts, issues, arguments, holdings, statutes cited, and precedents cited — so that a holding and its qualifying exceptions are not separated across chunk boundaries. Each chunk is enriched with structured metadata (case name, court, date, paragraph/page number, and citation identifiers) that is carried through indexing and retrieval so that every retrieved chunk can be traced back to its exact source location.

3.5 Feature Selection

The retrieval stage combines two complementary relevance signals — a lexical signal (BM25 term-weighted score) and a semantic signal (dense embedding cosine similarity) — together with hard metadata filters (for example, restricting by court or date range) applied prior to scoring. The relative weighting of the lexical and dense signals, and the number of candidates passed to the reranker, are treated as tunable hyperparameters (Section 4.3) rather than fixed a priori, so that the contribution of each signal can be evaluated empirically (Section 5.5, Section 5.7).

3.6 Proposed Model

The proposed model is a hybrid-retrieval, evidence-grounded generation system with four functional components: (i) a lexical retriever (BM25) over the indexed corpus; (ii) a dense retriever using sentence-embedding similarity search over a vector index (FAISS/ChromaDB); (iii) a cross-encoder reranker that re-scores the union of the top candidates from (i) and (ii) using joint query–passage attention for higher precision; and (iv) a generation-and-verification module in which an LLM produces a concise answer

conditioned only on the reranked top passages, with each claim in the answer linked to a specific supporting passage. A verification step checks whether each claim is entailed by its cited passage and issues a low-confidence / "insufficient evidence" response when support is weak, rather than allowing the system to answer regardless of evidence quality.

3.7 Baseline Models

- Baseline 1 — Keyword/Boolean search: representative of existing legal search websites; returns whole documents ranked by term match.
- Baseline 2 — BM25-only retrieval: lexical retrieval without dense or reranking components.
- Baseline 3 — Dense-only retrieval: semantic embedding retrieval without lexical signal or reranking.
- Baseline 4 — Unconstrained LLM (no retrieval): the query is answered directly by an LLM without any retrieved evidence, representing the hallucination-prone alternative this work aims to improve upon.

3.8 Explainability Method

Explainability is implemented at the claim level rather than the model-internal level: every generated claim is required to carry a citation to the case, paragraph/page, and passage from which it was derived. A faithfulness/entailment check compares each claim against its cited passage, and claims that are not well supported are flagged and either removed or replaced with an explicit statement that the evidence is insufficient. This design makes the explanation directly auditable by a human legal researcher, who can open the cited passage and verify the claim, rather than relying on an indirect, model-internal attribution method.

3.9 Algorithm / Mathematical Formulation

Hybrid retrieval score for a candidate passage d given query q:

$$Score(q, d) = \alpha \cdot BM25_norm(q, d) + \beta \cdot cos_sim(E(q), E(d)), \quad where\ \alpha + \beta = 1$$

Here, BM25_norm(q, d) is the min–max normalized BM25 score, E(·) denotes the dense sentence-embedding function, cos_sim is cosine similarity, and α, β are tunable weights (Section 4.3). The top-N candidates by Score(q, d) are passed to a cross-encoder reranker r(q, d) that outputs a refined relevance score used to select the final top-k passages supplied to the generator.

Retrieval evaluation metrics used to answer RQ1–RQ2:

$$Recall@k = |Relevant \cap Retrieved_k| \ / \ |Relevant|$$

$$Precision@k = |Relevant \cap Retrieved_k| \ / \ k$$

$$MRR = (1/|Q|) \cdot \Sigma\ (1 \ / \ rank_i), \text{ where } rank_i \text{ is the position of the first relevant passage for query } i$$

Faithfulness of a generated answer is computed as the proportion of claims that are entailed by their cited passage: Faithfulness = (number of supported claims) / (total number of claims). These quantities are used together, rather than in isolation, to answer RQ3 and RQ4 in Section 5.

4. EXPERIMENTAL SETUP

4.1 Hardware and Software

The prototype is designed to run on a single modern GPU or a cloud notebook environment, with retrieval kept local while the generation component remains modular and swappable. The software stack (Table 3) uses Python as the core implementation language, sentence-transformers for dense legal-domain embeddings, BM25 for lexical retrieval, FAISS/ChromaDB for vector indexing and similarity search, a cross-encoder model for reranking, an open-source or API-based LLM for grounded answer generation, and a dedicated evaluation suite for Recall@k, MRR, and faithfulness scoring.

Component	Tool / Technology	Role
Core language	Python	ML / NLP pipeline implementation
Dense embeddings	Sentence-Transformers	Legal-domain semantic representations
Lexical retrieval	BM25	Keyword-level retrieval signal
Vector index	FAISS / ChromaDB	Similarity search over dense embeddings
Reranking	Cross-Encoder Reranker	Re-orders top-k hybrid candidates
Generation	LLM (open-source / API)	Evidence-grounded answer generation
Ingestion	OCR & Text Cleaning	Judgment preprocessing
Evaluation	Evaluation Suite	Recall@k, MRR, faithfulness scoring

Table 3. Planned software stack for the retrieval-augmented legal assistant.

4.2 Dataset Split

Judgments are split at the case level (not the chunk level, to avoid leakage of passages from the same case across splits) into training/development and held-out test queries, with an additional time-based or jurisdiction-based split used to check that retrieval performance generalizes beyond the specific cases seen during development.

4.3 Hyperparameter Configuration

Key hyperparameters include the lexical/dense weighting (α, β) in the hybrid score, the number of candidates retrieved prior to reranking, the final top-k passed to the generator, the chunk size and overlap used in legal-structure-aware chunking, and the LLM decoding temperature (kept low to favor faithfulness over creativity). These are selected via grid search on the development split, optimizing primarily for Recall@k and faithfulness rather than fluency alone.

4.4 Cross-Validation

Given the case-level dependency between passages from the same judgment, standard k-fold cross-validation is applied at the case level (grouped cross-validation), ensuring that no two folds contain

passages from the same underlying judgment, which would otherwise inflate retrieval performance estimates.

4.5 Evaluation Metrics

Evaluation is organized along five dimensions (Table 4), each targeting a specific research question.

Dimension	Measures	Research Question
Retrieval	Recall@k, Precision@k, MRR / nDCG	Are relevant precedents/passages retrieved?
Answer quality	Rubric-based correctness and relevance	Does the response answer the legal question?
Faithfulness	Claim-to-source entailment / citation correctness	Are claims supported by retrieved evidence?
Usability	Latency, answer length, citation coverage	Is the assistant practical for research use?
Ablation	BM25 vs. dense vs. hybrid vs. hybrid+reranker	Which component contributes most to performance?

Table 4. Planned evaluation dimensions and corresponding research questions.

Target outcome: improve evidence retrieval and citation coverage over a simple keyword baseline while reducing the rate of unsupported generated claims.

4.6 Statistical Testing

Differences between the hybrid pipeline and each baseline (Section 3.7) on Recall@k, MRR, and faithfulness are planned to be assessed using paired significance tests appropriate for per-query comparisons — a paired t-test where score differences are approximately normally distributed, and a Wilcoxon signed-rank test as a non-parametric alternative — with significance assessed at $\alpha = 0.05$ and effect sizes reported alongside p-values so that statistical and practical significance can be distinguished.

5. RESULTS

Note on the status of results: *This project is reported at the design-and-prototype stage (see the ten-week timeline in Table 8 under Section 8 discussion of future work). The tables and figures below are therefore presented as an* ***illustrative walkthrough of the evaluation framework*** *described in Section 4 — using representative, non-final example values — so that the intended reporting format is fully specified in advance. They are not claimed as final experimental findings. Final measured values are to be substituted into these same tables once the planned experiments are executed.*

5.1 Overall Performance

Table 5 illustrates the intended top-line reporting format comparing the full proposed pipeline against the strongest baseline.

Metric	Keyword Baseline (illustrative)	Proposed Hybrid + Rerank (illustrative)
Recall@10	0.798	**0.940**
MRR	0.682	**0.914**
Faithfulness (%)	75.4%	**95.8%**
Median latency (s)	3.40s	**1.75s**

Table 5. Illustrative overall-performance reporting template (values pending experiment execution).

5.2 Model Comparison

The comparison across configurations is intended to be reported in ascending order of pipeline complexity — Baseline 1 (Boolean/keyword), Baseline 2 (BM25-only), Baseline 3 (dense-only), and the proposed hybrid pipeline with and without the cross-encoder reranker — using the metrics in Table 4, so that the marginal contribution of hybridization and of reranking can each be isolated.

5.3 Confusion Matrix

Because the system does not perform multi-class classification, the confusion-matrix analysis is repurposed here to describe the faithfulness-verification module, which performs a binary decision for each generated claim: "supported by cited evidence" versus "not adequately supported." Table 6 shows the intended reporting layout — a confusion matrix comparing the verification module's decision against a human-annotated gold label for a sample of generated claims.

	Gold: Supported	Gold: Not Supported
Predicted: Supported	True Positive (count TBD)	False Positive (count TBD)
Predicted: Not Supported / Abstain	False Negative (count TBD)	True Negative (count TBD)

Table 6. Planned confusion matrix for the claim-level faithfulness-verification decision.

5.4 ROC/PR Analysis

The faithfulness/entailment score used to trigger the abstain response is a continuous confidence value, so a Receiver Operating Characteristic (ROC) curve and a Precision–Recall (PR) curve are planned across a

sweep of abstain-threshold values, in order to select an operating threshold that balances the cost of withholding a correct answer (false abstain) against the cost of presenting an unsupported claim (false acceptance). Given the asymmetric cost of these two error types in a legal setting, the PR curve — and specifically precision at high recall — is expected to be the more decision-relevant view rather than the ROC curve alone.

5.5 Feature Importance

"Feature importance" here corresponds to the relative contribution of each retrieval signal to final answer quality, planned to be estimated by an ablation-based sensitivity analysis: measuring the drop in Recall@k and faithfulness when the lexical (BM25) signal, the dense signal, or the reranker is individually removed from the hybrid pipeline (Section 3.5, Section 4.6), rather than by an internal model feature-attribution method such as SHAP or LIME, which is less directly interpretable for a retrieval pipeline.

5.6 Explainability Results

Explainability is assessed by citation coverage — the proportion of generated claims that carry an attached case/paragraph/citation reference — and by human spot-checking of whether the cited passage actually entails the claim. A qualitative example format is planned in which a sample query, the retrieved passages, the generated answer, and the attached citations are shown side by side, illustrating the intended claim-to-evidence traceability described in Section 3.8.

5.7 Ablation Study

Table 7 shows the planned ablation design; each row removes exactly one component from the full pipeline.

Configuration	Components Removed	Expected Effect (to be measured)
Full pipeline	None	Reference configuration
– Reranker	Cross-encoder reranker	Expected drop in Precision@k
– Dense signal	Dense embedding retrieval	Expected drop in Recall@k for semantically-phrased queries
– Lexical signal	BM25 retrieval	Expected drop in Recall@k for citation/statute-exact queries
– Verification module	Faithfulness check / abstain	Expected increase in unsupported claims

Table 7. Planned ablation configurations.

5.8 Statistical Results

Consistent with Section 4.6, statistical results are planned to be reported as paired-test p-values (t-test and Wilcoxon signed-rank) comparing the full hybrid pipeline against each baseline on Recall@k, MRR, and faithfulness, together with effect sizes, once the experiments described in Section 4 have been executed.

6. DISCUSSION

6.1 Interpretation of Findings

The proposed architecture is designed so that each added component addresses a specific, named weakness of the baselines: hybrid retrieval addresses the recall limitations of keyword-only search; legal-structure-aware chunking addresses the fragmentation of holdings under fixed-size chunking; and the evidence-first generation-and-verification stage directly addresses the unsupported-claim problem of standalone LLMs. Once executed, the ablation study (Section 5.7) is expected to show that no single component is sufficient on its own — for example, lexical retrieval alone should perform comparatively well on exact citation or statute queries but poorly on semantically phrased queries, and vice versa for dense retrieval alone, motivating the hybrid design.

6.2 Comparison With Previous Studies

The design choices in this paper are consistent with, and extend, findings reported in prior work: the preference for precise, narrowly scoped passages over large chunks aligns with LegalBench-RAG (Pipitone & Houir Alami, 2024); the combination of realistic case context with precedent retrieval aligns with NyayaRAG (Nigam et al., 2025); and the explicit concern with unsupported claims in AI legal research tools is directly motivated by the empirical finding of continued hallucination in leading commercial legal AI tools reported by Magesh et al. (2024). This paper's contribution relative to that prior work is the combination of hybrid retrieval, legal-structure-aware chunking, and claim-level faithfulness verification within a single evaluated pipeline, together with a five-dimension evaluation protocol that reports these aspects separately rather than as one blended score.

6.3 Theoretical Implications

The work suggests that, for high-stakes, citation-dependent domains such as law, retrieval quality and generation faithfulness should be treated as jointly necessary but individually insufficient conditions for a trustworthy system — a high-recall retriever paired with an ungrounded generator is expected to remain unreliable, and a well-grounded generator paired with a low-recall retriever cannot answer questions correctly regardless of how faithfully it cites what little evidence it received.

6.4 Practical Implications

For practicing legal researchers, students, and clerks, a system that returns a short, cited, confidence-scored answer rather than a list of documents to read individually could materially reduce the time spent on preliminary precedent search, while the abstain mechanism is intended to preserve the requirement that a human ultimately verifies and exercises judgment over any output used in practice.

6.5 Unexpected Findings

As the quantitative experiments have not yet been executed, no unexpected empirical findings can be reported at this stage. One anticipated area of interest, to be examined once results are available, is whether the reranker's marginal contribution differs systematically between short factual queries and broader "which precedents support X" style queries, since these two query types are expected to depend differently on lexical versus semantic matching.

7. LIMITATIONS

- The evaluation reported in Section 5 is illustrative of the intended reporting format; final quantitative results have not yet been produced, pending completion of the experiments in Section 4.
- The initial corpus is scoped to a single jurisdiction/court, which limits the generalizability of findings to other legal systems or languages until the corpus is extended.
- Faithfulness/entailment scoring and rubric-based answer-quality judgments partly rely on human annotation, which is time-consuming to produce at scale and may introduce annotator subjectivity.
- The system is explicitly a research-support prototype and is not designed, validated, or intended for autonomous legal decision-making or as a substitute for qualified legal advice.
- Latency and compute constraints of running a cross-encoder reranker and an LLM in sequence may limit real-time usability at larger corpus scale without further optimization.

8. CONCLUSION AND FUTURE WORK

This paper has proposed a retrieval-augmented legal assistant for court-judgment question answering that combines hybrid lexical–dense retrieval with cross-encoder reranking, legal-structure-aware chunking, and evidence-first, citation-linked generation with an explicit faithfulness-verification and abstain mechanism. The intended outcome is a prototype legal research assistant that retrieves relevant court-judgment passages, summarizes evidence in plain language, cites source passages and cases, exposes its own confidence and uncertainty, and supports rather than replaces human legal research. Beyond its applied outcome, the project demonstrates applied machine learning for text analytics through retrieval, semantic similarity, reranking, generation, evaluation, and error analysis in an integrated pipeline.

The remaining work is organized as a ten-week schedule (Table 8), moving from scoping and literature survey through to a final evaluation, error analysis, and demonstration.

Week	Planned Activity
W1	Scope definition and literature survey
W2	Dataset collection and metadata schema design
W3	Cleaning, OCR checks, and legal-aware chunking
W4	BM25 and dense retrieval baseline implementation
W5	Hybrid retrieval and reranker integration
W6	RAG prompt design and citation formatting
W7	Faithfulness / confidence verification module
W8	Full evaluation and ablation experiments
W9	Error analysis and UI/prototype refinement
W10	Final report, presentation, and demonstration

Table 8. Planned ten-week project schedule, with milestones at Week 4 (working retrieval baseline), Week 7 (grounded RAG prototype), and Week 10 (evaluation and final demonstration).

Future work beyond the current scope includes extending the corpus to additional jurisdictions and languages, incorporating expert legal review of a larger annotated sample for faithfulness evaluation, and exploring whether structured legal reasoning (for example, explicit statute-to-holding linkage) can further improve both retrieval precision and the interpretability of generated answers.

DECLARATIONS

Funding

This work was carried out as part of an academic course project (Applied Machine Learning for Text Analytics) and did not receive external or dedicated research funding.

Conflict of Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethics Statement

The proposed system uses publicly accessible or institution-provided court-judgment documents and does not process private personal data beyond what is already contained in public judgment records. The system is explicitly positioned as a decision-support research tool and is not intended to provide legal advice or to be used for autonomous legal decision-making; outputs require human verification before any practical use.

Data Availability

The judgment corpus used for development is drawn from publicly accessible repositories or institution-provided datasets. Access to the specific dataset compiled for this project is available from the corresponding author on reasonable request, subject to the terms of use of the original data source.

Author Contributions

A. Sahasra: conceptualization, system architecture, manuscript writing. K. P. Shreya: literature survey, retrieval pipeline design. V. Gowrie: data collection and preprocessing design, evaluation framework. K. Ishta: methodology formalization, project planning, and manuscript review. All authors read and approved the final manuscript.

ACKNOWLEDGEMENTS

The authors thank their course instructor and peers in Applied Machine Learning for Text Analytics for feedback on the project proposal and system design.

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Note: This reference list contains 29 entries — the six sources identified directly in the project literature survey plus well-established foundational works in information retrieval, retrieval-augmented generation, and legal NLP that support the methodological claims made in this paper. The target style guide requests at least 30 references; readers preparing this paper for actual journal submission should expand this list with additional recent, closely-related legal-RAG and legal-NLP papers identified through their own literature search, and should verify every citation against the original source before submission.